8	(a)	discrete quantity / packet / quantum of energy of electromagnetic radiation		B1	
		energy of photon = Planck constant × frequency		B1	[2]
	(b)	threshold frequency	(1)		
		rate of emission is proportional to intensity	(1)		
		max. kinetic energy of electron dependent on frequency	(1)		
		max. kinetic energy independent of intensity	(1)		
		(*any three, 1 each, max 3*)		B3	[3]

(c) *either* $E = hc/\lambda$	*or* $hc/\lambda = eV$	C1	
λ = 450 nm to give	work function of *3.5 eV*		
energy = 4.4×10^{-19} or 2.8 eV	to give λ = 355 nm	M1	
2.8 eV < 3.5 eV so no emission	355 nm < 450 nm so no	A1	[3]

or work function = 3.5 eV

threshold frequency = 8.45×10^{14} Hz	C1	
450 nm = 6.67×10^{14} Hz	M1	
6.67×10^{14} Hz < 8.45×10^{14} Hz	A1	

Section B

9 (a) e.g. zero output impedance / resistance